

6. The photograph shows the skull of a wild boar (*Sus scrofa*).



(a) Like humans, wild boars are omnivores. From the photograph describe **one** feature of the wild boar's dentition that is typical of a carnivore's dentition and **one** feature that is typical of a herbivore's dentition. For each feature explain how it is suited to the diet of the wild boar.

(i) Carnivore. [1]

.....

.....

.....

.....

(ii) Herbivore. [1]

.....

.....

.....

.....



(b) The alimentary canal of a wild boar is very similar to that of a human. **State the region** of the human alimentary canal where the digestion of the following dietary constituents **begins:** [2]

- Lipids
- Starch
- Proteins

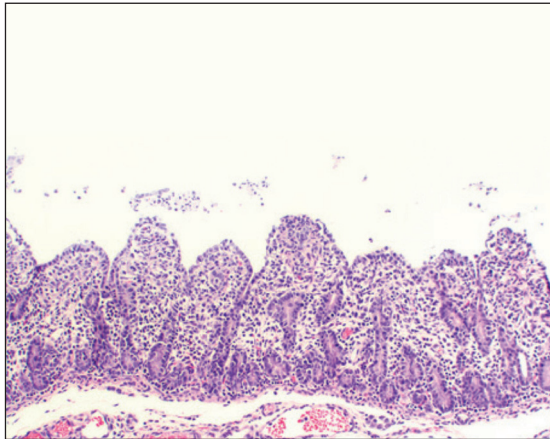
(c) Porcine epidemic diarrhoea (PED) is a condition that is caused by a virus and leads to severe gastrointestinal disease in wild boars. The virus infects and destroys epithelial cells at the tips of villi in the small intestine. Symptoms include weight loss and diarrhoea.

The photomicrograph shows the lining of the small intestine of a normal wild boar and of a wild boar suffering with PED.

Normal wild boar



Wild boar with PED



With reference to the photomicrographs and your own knowledge, describe and explain how the viral infection causes weight loss and diarrhoea. [5]

.....

.....

.....

.....

.....

.....

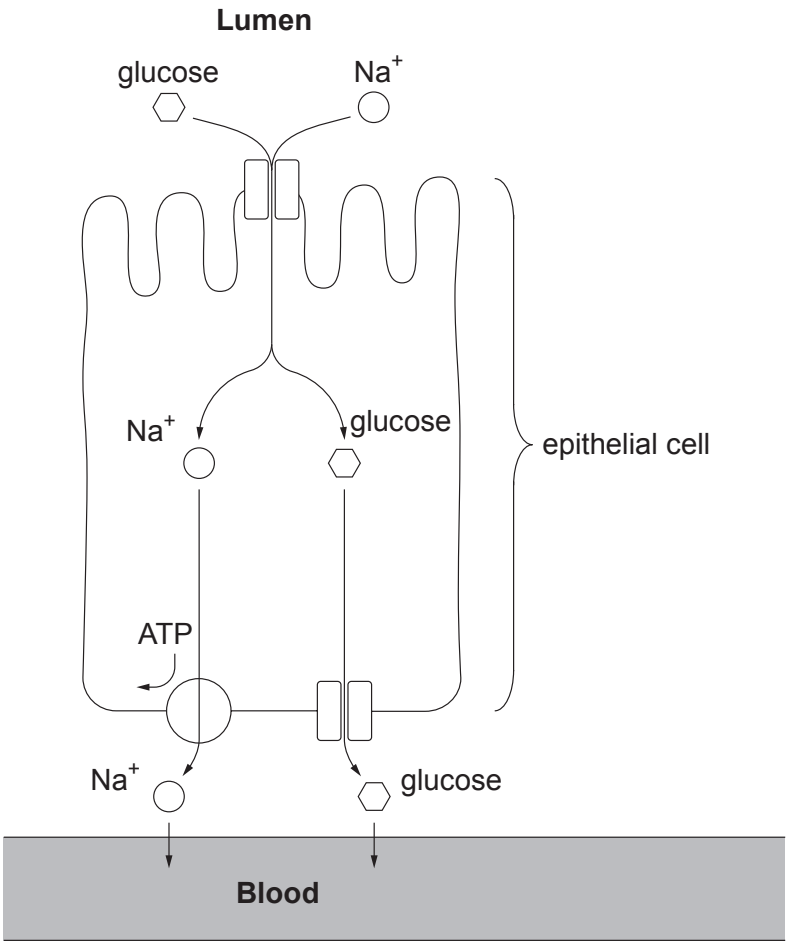
.....

.....



- (d) The epithelial cells of villi contain many mitochondria. These supply ATP for the active transport of sodium ions (Na^+) out of epithelial cells and eventually into blood capillaries.

Use the diagram below and your own knowledge to explain how the active transport of sodium ions out of the epithelial cell results in passage of glucose from the lumen of the small intestine, across the epithelium and into the bloodstream. [4]



.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

